Group 7 Summary:Smoking Behavior & BMI Analysis

1 Research Questions

- **Is the proportion of students who have ever smoked different from 50%?**
- **Is the mean BMI percentile different from 65?**

2 Data

data:YRBS_2007
Behavior Variable:EverCigaretteUse
Continuous Variable:BMIPCT

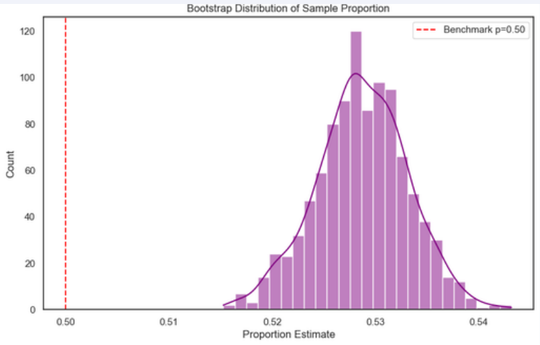
3 EDA、Method

1.Behavioral Variable:Distribution of Past Smoking Behavior

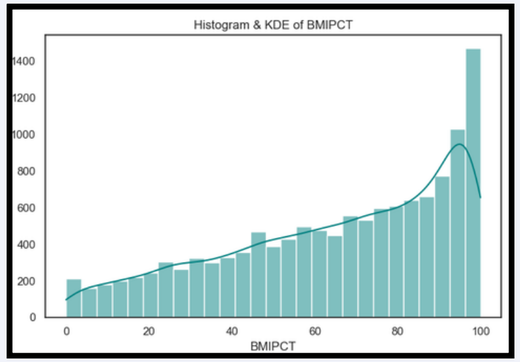
Analysis Objective:To observe the proportion of "past smokers" and "never smokers" in the sample using bar charts, and to calculate the sample proportion.

2.Continuous Variable: BMI Percentile Distribution (BMIPCT)

Analysis Objective: To examine the shape of the BMI percentile distribution. This helps confirm whether the data meets the assumptions of a single-sample t-test.

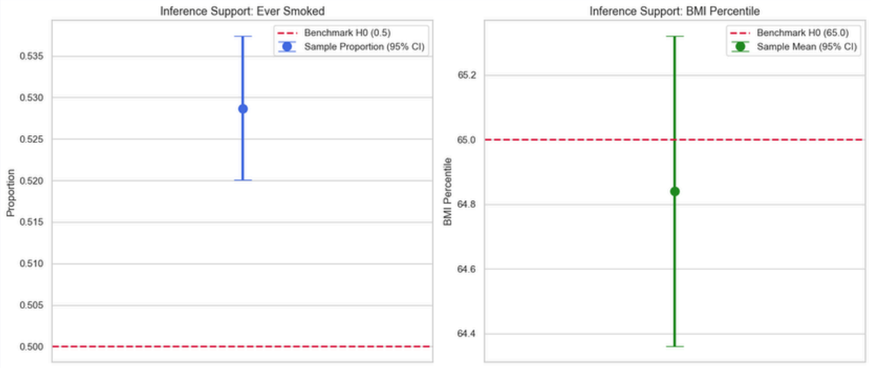


EverCigaretteUse



BMIPCT

4 Hypothesis Testing



Smoking Proportion Test

Sample proportion = 52.87%
p-value = < 0.001
95% CI = (52.01%, 53.74%)
Interpretation:
The proportion of students who have ever smoked is significantly higher than 50%.

BMI Mean Test

Mean = 65
p-value = 0.5191
95% CI = (64.36, 65.32)
Interpretation:
The mean BMI percentile is not significantly different from 65.

5 Final Conclusion

The EDA and hypothesis testing consistently show that:

Smoking prevalence among adolescents in 2007 was above half of the sample
Average BMI percentile remained close to the expected benchmark
Although BMI values vary widely, the overall mean is stable

These findings suggest that smoking was relatively common, while BMI levels remained within the normal expected range.